

BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```

Activating project at `~/Desktop/Cornell_Courses/FA_25/BEE_4750/hw2-xindiliu`
Installed InvertedIndices          v1.3.1
Installed JpegTurbo_jll            v3.1.3+0
Installed WorkerUtilities           v1.6.1
Installed SentinelArrays            v1.4.8
Installed InlineStrings             v1.4.5
Installed Accessors                 v0.1.42
Installed Roots                    v2.2.10
Installed DataFrames                v1.8.0
Installed WeakRefStrings            v1.4.2
Installed PooledArrays              v1.4.3
Installed ConstructionBase          v1.6.0
Installed DataValueInterfaces       v1.0.0
Installed Ghostscript_jll           v9.55.1+0
Installed TableTraits               v1.0.1
Installed CompositionsBase          v0.1.2
Installed InverseFunctions          v0.1.17
Installed CommonSolve               v0.2.4
Installed IteratorInterfaceExtensions v1.0.0
Installed Crayons                   v4.1.1
Installed Tables                    v1.12.1
Installed Latexify                  v0.16.10
Installed PrettyTables              v3.0.8
Installed Glib_jll                  v2.86.0+0
Installed FilePathsBase             v0.9.24
Installed StringManipulation        v0.4.1
Installed CSV                       v0.10.15
Precompiling project...
 357.9 ms  IteratorInterfaceExtensions
 319.3 ms  CommonSolve
 326.4 ms  DataValueInterfaces
 379.5 ms  WorkerUtilities
 400.2 ms  InverseFunctions
 281.8 ms  CompositionsBase
 283.8 ms  InvertedIndices
 325.2 ms  ConstructionBase
 486.7 ms  InlineStrings
 396.8 ms  PooledArrays
 742.8 ms  Crayons
 486.7 ms  JpegTurbo_jll
 889.7 ms  SentinelArrays
 636.9 ms  FilePathsBase
 701.0 ms  Glib_jll

```

1279.7 ms	StringManipulation
561.6 ms	TableTraits
418.6 ms	InverseFunctions → InverseFunctionsDatesExt
350.7 ms	LogExpFunctions → LogExpFunctionsInverseFunctionsExt
573.6 ms	InverseFunctions → InverseFunctionsTestExt
542.8 ms	Unitful → InverseFunctionsUnitfulExt
395.6 ms	CompositionsBase → CompositionsBaseInverseFunctionsExt
485.7 ms	ConstructionBase → ConstructionBaseLinearAlgebraExt
482.0 ms	Unitful → ConstructionBaseUnitfulExt
512.9 ms	InlineStrings → ParsersExt
529.1 ms	FilePathsBase → FilePathsBaseMmapExt
647.3 ms	FilePathsBase → FilePathsBaseTestExt
3007.8 ms	ColorSchemes
1428.4 ms	Libtiff_jll
1493.1 ms	Ghostscript_jll
1052.3 ms	Tables
813.9 ms	WeakRefStrings
1765.7 ms	Cairo_jll
2341.0 ms	Qt6Base_jll
2064.5 ms	Accessors
868.4 ms	HarfBuzz_jll
2490.8 ms	Latexify
528.1 ms	Accessors → LinearAlgebraExt
1084.1 ms	Qt6ShaderTools_jll
422.5 ms	Accessors → TestExt
484.7 ms	Accessors → UnitfulExt
762.5 ms	libass_jll
704.7 ms	Pango_jll
791.8 ms	Latexify → SparseArraysExt
895.5 ms	UnitfulLatexify
6357.7 ms	PlotUtils
1781.5 ms	Qt6Declarative_jll
1342.3 ms	FFMPEG_jll
2769.4 ms	Roots
1056.2 ms	FFMPEG
2132.8 ms	PlotThemes
798.9 ms	Roots → RootsUnitfulExt
1008.8 ms	GR_jll
2893.7 ms	RecipesPipeline
10508.1 ms	CSV
2422.1 ms	GR
19333.2 ms	PrettyTables
24988.0 ms	DataFrames

```
1431.9 ms    Latexify → DataFramesExt
33973.5 ms   Plots
2844.7 ms    Plots → UnitfulExt
61 dependencies successfully precompiled in 55 seconds. 158 already precompiled.
```

```
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO₂ concentrations, and ocean CO₂ concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

Tip

Think about Henry's law for CO₂.

Carbon Cycle and Mean Temperature Systems Diagram

The interaction between atmospheric CO₂ concentrations and global mean temperature is characterized by a positive feedback loop. An increase in atmospheric CO₂ enhances the greenhouse effect, leading to a rise in global mean temperature, which is also why it is a positive relationship. This warming, in turn, reduces the ocean's capacity to store CO₂, as the solubility of gases in water decreases with increasing temperature according to Henry's Law, so it is shown by a negative relationship from temperature to ocean CO₂ storage. Consequently, with the ocean's carbon sink efficiency diminished, more CO₂ is released into and remains in the atmosphere; again shown by a negative relationship from ocean CO₂ storage to atmospheric CO₂. This chain of effects creates a self-amplifying cycle: higher CO₂ levels cause warming, which reduces oceanic uptake, leading to even higher atmospheric CO₂ levels and further warming. This critical feedback loop accelerates climate change and is a fundamental concept in climate science.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/out-flows between these boxes. How did you decide how many boxes were needed?

River Control Volume System Diagram

To represent the river system, I divided it into two control volumes. or boxes, corresponding to the two distinct river reaches between point source discharges.

Box 1: The segment of river from the first mixing point at $x = 0$, where the upstream flow and the first wastewater discharge mix, to the second mixing point at $x = 15 \text{ km}$.

Box 2: The segment of river downstream of the second mixing point ($x = 15 \text{ km}$), which includes the combined flows from Box 1 and the second wastewater discharge.

I decided to use two boxes based on the structure of the problem: between $x = 0$ and $x = 15 \text{ km}$ the river is subject only to the first discharge and distributed processes, while after $x = 15 \text{ km}$ the concentration is reset by mixing with the second discharge, creating a new starting condition for downstream transport. If the river is treated as one box only, the additional source introduced at $x = 15 \text{ km}$ is not captured. On the other hand, using more than two boxes is unnecessary because there are only two discrete points where the mass balance is changed.

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

Tip

Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Continuous-time CRUD transport model (in distance x):

$$\frac{dC}{dx} = -\frac{k}{v}C + \frac{r_{\text{dep}}}{Q},$$

where C is concentration, k is the decay rate, v is the river velocity, r_{dep} is the deposition input, and Q is discharge.

Forward Euler approximation (step size Δx):

$$\frac{dC}{dx} \approx \frac{C_{n+1} - C_n}{\Delta x}.$$

Substitute into the ODE:

$$\frac{C_{n+1} - C_n}{\Delta x} = -\frac{k}{v}C_n + \frac{r_{\text{dep}}}{Q}.$$

Solve for C_{n+1} :

$$C_{n+1} = C_n + \Delta x \left(-\frac{k}{v}C_n + \frac{r_{\text{dep}}}{Q} \right).$$

For $\Delta x = 1$ km:

$$C_{n+1} = \left(1 - \frac{k}{v} \right) C_n + \frac{r_{\text{dep}}}{Q}.$$

Problem 2.3

Determine if the system is in compliance with a regulatory limit of 2.3 kg/(1000m³). You can do this analytically or computationally.

On reach 1 ($0 \leq x \leq 15$ km):

$$C_1(x) = C_1(0) e^{-ax} + P_1(1 - e^{-ax}),$$

where

$$\begin{aligned} a &= \frac{k}{v} = \frac{0.36}{10} = 0.036 \text{ km}^{-1}, \\ C_1(0) &= 1.672 \text{ kg}/(1000 \text{ m}^3), \\ P_1 &= \frac{r_{\text{dep}} v}{k Q_1} \cdot 1000 = \frac{54 \cdot 10}{0.36 \cdot 290000} \cdot 1000 \approx 5.17 \text{ kg}/(1000 \text{ m}^3). \end{aligned}$$

Set $C_1(x) = 2.3$ and solve for x :

$$e^{-ax} = \frac{2.3 - P_1}{C_1(0) - P_1}.$$

Numerically,

$$\frac{2.3 - P_1}{C_1(0) - P_1} = \frac{2.3 - 5.172}{1.672 - 5.172} \approx 0.8207,$$

so

$$x = -\frac{1}{a} \ln(0.8207) = -\frac{1}{0.036} \ln(0.8207) \approx 5.49 \text{ km}.$$

Thus on Reach 1 the concentration exceeds the regulatory limit of 2.3 kg/(1000 m³) for $x \gtrsim 5.5$ km (i.e. between about 5.5 km and the mixing at 15 km).

At the mixing at $x = 15$ km:

$$\begin{aligned} C_1(15) &\approx 3.13 \text{ kg/(1000 m}^3\text{)}, \\ C_2(0) &\approx 3.80 \text{ kg/(1000 m}^3\text{)}. \end{aligned}$$

On Reach 2 concentrations remain above the limit and asymptote to

$$C_\infty \approx 4.29 \text{ kg/(1000 m}^3\text{)}.$$

Conclusion: The system is not in compliance with the regulatory limit. Concentrations exceed 2.3 kg/(1000 m³) from roughly $x \approx 5.5$ km downstream of the first mixing point and remain above the limit after the second discharge.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized “Snowball Earth”. In this problem, we’ll introduce a simple model of the Earth’s energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C \frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a \ln \left(\frac{[CO_2]}{[CO_2]_{PI}} \right)}_{\text{greenhouse effect}}, \quad (1)$$

where:

- T is the Earth’s global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be $51 \text{ J/m}^2/\text{°C}$;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);
- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3 \text{ W/m}^2/\text{°C}$ (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2 \text{ W/m}^2$ (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1 \text{ yr}$.

Continuous-time climate model (ignoring greenhouse effect):

$$C \frac{dT}{dt} = \frac{(1-\alpha)S}{4} - (A - BT)$$

Forward Euler approximation:

$$\frac{dT}{dt} \approx \frac{T_{n+1} - T_n}{\Delta t}$$

Substitute into the ODE:

$$C \frac{T_{n+1} - T_n}{\Delta t} = \frac{(1 - \alpha)S}{4} - (A - BT_n)$$

Solve for T_{n+1} :

$$T_{n+1} = T_n + \frac{\Delta t}{C} \left(\frac{(1 - \alpha)S}{4} - A + BT_n \right)$$

For $\Delta t = 1$ year:

$$T_{n+1} = \left(1 + \frac{B}{C} \right) T_n + \frac{1}{C} \left(\frac{(1 - \alpha)S}{4} - A \right)$$

Numerical values:

$$C = 51 \text{ J m}^{-2} \text{ }^\circ\text{C}^{-1}, \quad B = -1.3 \text{ W m}^{-2} \text{ }^\circ\text{C}^{-1}$$

Discrete update formula:

$$T_{n+1} \approx 0.9745 T_n + 0.0196 \left(\frac{(1 - \alpha)S}{4} - A \right)$$

Problem 3.2

Rather than assuming a constant value for the albedo α , we will represent the ice-albedo feedback by letting α depend on T :

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10^\circ\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10^\circ\text{C} \leq T \leq 10^\circ\text{C} \\ \alpha_0 & \text{if } T \geq 10^\circ\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoproterozoic value of S with temperature-varying albedo for initial values of T spanning $-60^\circ\text{C} \leq T_0 \leq 30^\circ\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```

# Parameters
C = 51.0          # Heat capacity (J/m2/°C)
A = 221.2         # Outgoing radiation parameter
B = -1.3          # Outgoing radiation sensitivity (W/m2/°C)
S = 1272.0        # Neoproterozoic solar constant (W/m2)
alpha_i = 0.5     # Ice-covered albedo
alpha_0 = 0.3     # Ice-free albedo
Δt = 1.0          # timestep in years
timesteps = 200   # number of years

# Albedo as a function of temperature
if T <= -10
    return alpha_i
elseif T >= 10
    return alpha_0
else
    return alpha_i + (alpha_0 - alpha_i) * (T + 10)/20
end
end

# Forward Euler update function
function update(T)
    return T + ( (1 - alpha(T))*S/4 - (A - B*T) ) * Δt / C
end

# Initial temperatures
T0 = range(-60, stop = 30, length = 50)

# Simulate trajectories
trajectories = []
for T_init in T0
    T = T_init
    traj = [T]
    for n in 1:timesteps
        T = update(T)
        push!(traj, T)
    end
    push!(trajectories, traj)
end

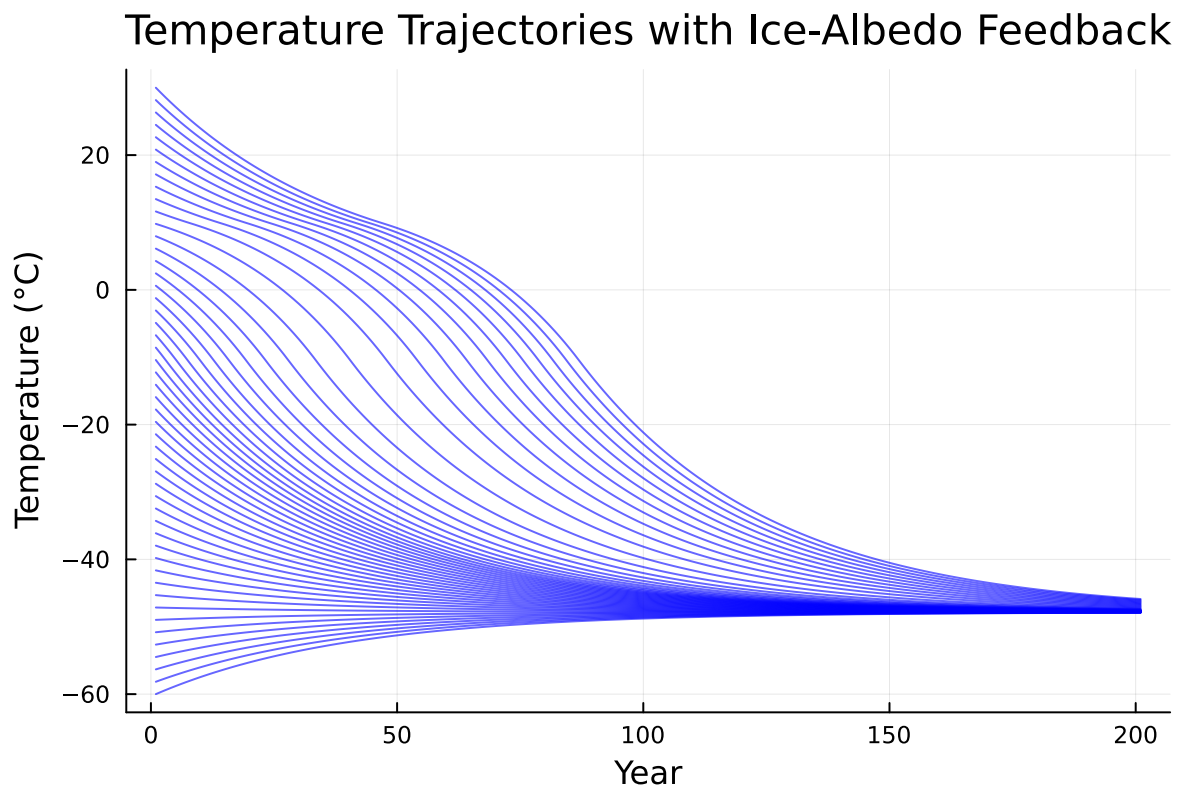
# Plot trajectories
using Plots

```

```

plot()
for traj in trajectories
    plot!(traj, color = :blue, alpha = 0.6, label = false)
end
xlabel!("Year")
ylabel!("Temperature (°C)")
title!("Temperature Trajectories with Ice-Albedo Feedback")

```



There is only one stable equilibrium at around -45°C , corresponding to the “Snowball Earth” state during the Neoproterozoic era. The solar constant $S = 1272 \text{ W/m}^2$ is lower than the present-day value of 1368 W/m^2 . This reduces the incoming solar radiation, making it harder for the Earth to escape a frozen state. The albedo function $\alpha(T)$ increases significantly when the temperature drops below -10°C , reflecting more sunlight and further cooling the planet. This positive feedback reinforces the cold state. All these reasons contribute to the stable equilibrium.

Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C ?

At equilibrium, the forward Euler climate model satisfies

$$0 = \frac{(1 - \alpha(T))S}{4} - (A - BT),$$

where T is temperature in $^\circ\text{C}$, S is the solar constant, A and B are outgoing radiation parameters, and $\alpha(T)$ is the temperature-dependent albedo. Solving for T gives

$$T = \frac{A - (1 - \alpha(T))S/4}{B}.$$

For the warm state (ice-free, $\alpha \approx \alpha_0 = 0.3$) using the Neoproterozoic solar constant $S = 1272 \text{ W m}^{-2}$:

$$\begin{aligned} T_{\text{warm}} &= \frac{A - (1 - \alpha_0)S/4}{B} \\ &= \frac{221.2 - 0.7 \times 1272/4}{-1.3} \\ &= \frac{221.2 - 222.6}{-1.3} \\ &= \frac{-1.4}{-1.3} \\ &\approx 1.08^\circ\text{C}. \end{aligned}$$

To reach the pre-industrial temperature $T = 14^\circ\text{C}$, the required solar constant would be

$$S = \frac{4(A - BT)}{1 - \alpha_0} = \frac{4(221.2 - (-1.3)(14))}{0.7} = \frac{957.6}{0.7} \approx 1368 \text{ W m}^{-2}.$$

Conclusion: The Neoproterozoic solar constant (1272 W m^{-2}) was too weak to give 14°C . However, the gradual increase of the solar constant over Earth's history (to ~ 1368 today) is, in principle, sufficient to allow such a transition. In practice, additional processes (like greenhouse gas buildup) were likely also involved, since solar luminosity increases on billion-year timescales, while Snowball Earth episodes ended much faster (millions of years).

References

List any external references consulted, including classmates.

1. Used ChatGPT for problems 2.2, 2.3, mainly for helping understand how the coding language works and what questions are asking, not for discussions and analysis of problems. OpenAI. (2025). ChatGPT (Feb 1 GPT-4 model) [Large language model]. <https://chat.openai.com>
2. Used DeepSeek for problems 3.2, 3.3, for similar purposes. DeepSeek. (2025). DeepSeek Chat (Jun 25, 2024 version) [Large language model]. <https://chat.deepseek.com>